

ladder. [A1]

2. (Due to the person's weight) there is now a greater anticlockwise moment about the ladder bottom, and so (to maintain equilibrium) the wall exerts a greater clockwise moment and hence greater horizontal force. [A1]

2 (a) uncertainty in $\Delta\theta = \frac{10.5 - 9.7}{2} = 0.4 \text{ K}$

OR Accept:

$$\text{average } \Delta\theta = \frac{10.2 + 9.7 + 10.5}{3} = 10.13 \text{ K}$$

$$\text{uncertainty in } \Delta\theta = 10.13 - 9.7 = 0.433 = 0.4 \text{ K}$$

(b)

$$\begin{aligned} c &= \frac{Q}{m\theta} = \frac{IVt}{m\theta} \\ &= \frac{4.125 \times 11.8 \times 200.0}{0.309 \times 10.13} && [\text{C1}] \\ &= 3018.2 && [\text{A1}] \end{aligned}$$

(c)

$$\begin{aligned}\frac{\Delta c}{c} &= \frac{\Delta V}{V} + \frac{\Delta l}{l} + \frac{\Delta t}{t} + \frac{\Delta m}{m} + \frac{\Delta(\theta)}{\theta} \\ &= \frac{0.3}{11.8} + \frac{0.002}{4.125} + \frac{0.5}{200.0} + \frac{3}{309} + \frac{0.4}{10.1} \quad [\text{M1}] \\ &= 0.077721 \\ \Delta c &= 3018.2 \times 0.077721 = 200 \quad [\text{M1}] \\ c &= 3000 \pm 200 \text{ J kg}^{-1} \text{ K}^{-1} \quad [\text{A1}]\end{aligned}$$

(d) No heat loss to surrounding. [B1]

3. (a) Newton's second law of motion states that the rate of change of momentum is proportional